

10.1



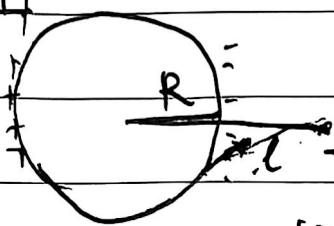
$$\varphi_a = \varphi_b = \frac{1}{4\pi\epsilon_0} \frac{Q+q}{b}$$

$$\varphi_0 - \varphi_a = \frac{1}{4\pi\epsilon_0} \frac{q}{r} - \frac{1}{4\pi\epsilon_0} \frac{q}{a-r} = \frac{Q+q}{4\pi\epsilon_0 b}$$

$$\therefore \varphi_0 = \left(\frac{1}{4\pi\epsilon_0} \left(\frac{Q+q}{b} + \frac{q}{r} - \frac{q}{a-r} \right) \right)$$

虽然电荷分布不均匀, 但实为积分:

10.4



$$(1) \varphi_{内} = \varphi_R = \varphi_0 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

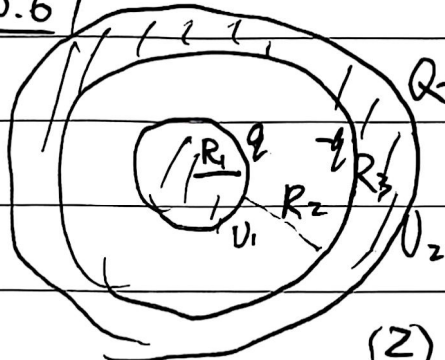
$$\varphi_0 = \frac{1}{4\pi\epsilon_0} \frac{q}{r} + \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

$$\therefore \varphi_{内} = \varphi_0 = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} + \frac{Q}{R} \right)$$

$$(2) \varphi_0 = 0 \therefore \frac{q}{r} + \frac{Q}{R} = 0$$

$$\Rightarrow q' = -\frac{Q}{R} r \quad \varphi = \frac{1}{4\pi\epsilon_0} \frac{q}{r} + \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = 0$$

10.6



$$(1) U_2 = \frac{1}{4\pi\epsilon_0} \frac{Q+q}{R_3}$$

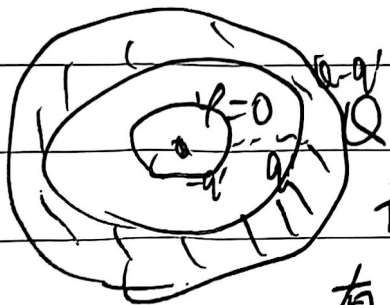
$$U_1 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R_1} - \frac{1}{4\pi\epsilon_0} \frac{q}{R_2} + \frac{1}{4\pi\epsilon_0} \frac{Q+q}{R_3}$$

$$\Rightarrow U_1 = \frac{1}{4\pi\epsilon_0} \left(\frac{Q+q}{R_3} + \frac{q}{R_1} - \frac{q}{R_2} \right)$$

$$(2) U_1 = U_2 = \frac{1}{4\pi\epsilon_0} \frac{Q+q}{R_3}$$

$$(3) U_1 = 0$$

$$\frac{Q}{R_1} - \frac{q}{R_2} + \frac{Q+q}{R_3} = 0$$



$$U_2 = U_1 = 0$$

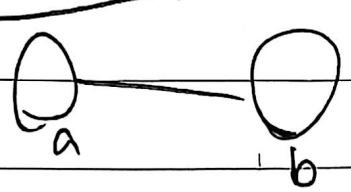
$$U_2 = 0 \therefore U_2 = U_1 = 0$$

设内球 $-q'$, 外球内表面 q' , 外表面 $Q-q'$

$$\text{有 } U_1 = \frac{1}{4\pi\epsilon_0} \left(\frac{Q-q'}{R_3} + \frac{q'}{R_2} - \frac{q'}{R_1} \right) = 0 \quad \frac{Q}{R_3} = q' \left(\frac{1}{R_3} + \frac{1}{R_2} - \frac{1}{R_1} \right)$$

$$\therefore U_2 = \frac{1}{4\pi\epsilon_0} \left(\frac{Q-q'}{R_3} \right) = \frac{Q}{4\pi\epsilon_0} \frac{R_2 - R_1}{R_1 R_2 + R_1 R_3 - R_2 R_3}$$

10.11



(1) 独立 $\Phi_a = \Phi_b$ $\varphi_a = \varphi_b$

设 q_a, q_b $q_a + q_b = Q$

有: $\varphi_a = \frac{1}{4\pi\epsilon_0} \frac{q_a}{a} = \varphi_b = \frac{1}{4\pi\epsilon_0} \frac{q_b}{b} \Rightarrow \begin{cases} q_a = \frac{a}{a+b} Q \\ q_b = \frac{b}{a+b} Q \end{cases}$

(2) 法1: 串联

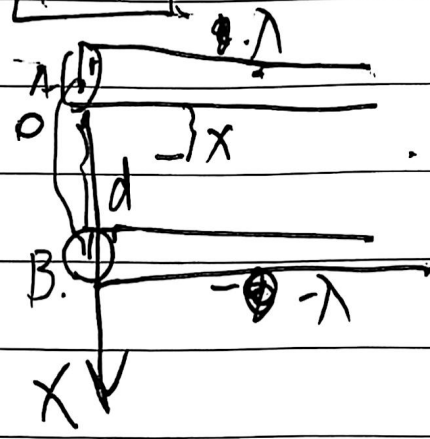
~~$C_a = \frac{q_a}{\varphi_a} = \frac{\frac{a}{a+b} Q}{\frac{1}{4\pi\epsilon_0} \frac{Q}{a+b}} = 4\pi\epsilon_0 a$~~

$C_b = 4\pi\epsilon_0 b$

$\therefore C = 4\pi\epsilon_0 (a+b)$

法2: $C = \frac{Q}{\varphi_a} = \frac{Q}{\frac{1}{4\pi\epsilon_0} \frac{Q}{a+b}} = 4\pi\epsilon_0 (a+b)$

10.13



~~独立~~ 设带电量 $\pm \lambda$

单位长度

如图建系

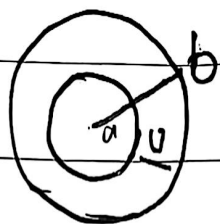
$-(\ln r - \ln(d-r))$

$E_x = \frac{\lambda}{2\epsilon_0 x} + \frac{\lambda}{2\epsilon_0 (d-x)}$

$\therefore U = \int_r^{d-r} E_x dx = \frac{\lambda}{2\epsilon_0} \int_r^{d-r} \frac{d}{x(d-x)} dx = \frac{\lambda}{2\epsilon_0} (\ln \frac{d}{r} + \ln \frac{d}{d-r})$
 $= \frac{\lambda}{\epsilon_0} \ln \frac{d}{r}$

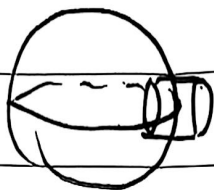
$\therefore C = \frac{Q}{U} = \lambda \epsilon_0 \frac{1}{\ln \frac{d}{r}}$

10.14



设 $\pm q, C = \frac{q}{U}$

内球:



取圆柱 + 高斯:

$$\oint \vec{E} \cdot d\vec{S} = E \cdot dS$$

$$\Rightarrow E = \frac{\sigma}{\epsilon_0}$$

$$\sigma = \frac{q}{4\pi a^2} \Rightarrow E = \frac{q}{4\pi a^2 \epsilon_0} = \frac{CU}{4\pi a^2 \epsilon_0}$$

已知: $C = 4\pi \epsilon_0 \frac{ab}{b-a}$

[推导: $U = \int_a^b E dr = \int_a^b \frac{1}{4\pi \epsilon_0} \frac{q}{r^2} dr = \frac{1}{4\pi \epsilon_0} (\frac{1}{b} - \frac{1}{a})$

$\therefore C = \frac{Q}{U} = 4\pi \epsilon_0 \frac{ab}{b-a}$

代入有: $E = \frac{ab}{a^2} U = \frac{b}{a} \cdot \frac{1}{b-a} U = \frac{b}{ab-a^2} U$

$\therefore$ 当 $a = \frac{b}{2}$ 时, $E_{min} = \frac{4}{b} U$

10.15 $S = 2 \times 10^{-3} m^2$ $d = 3 \times 10^{-3} m$

(1) 左介质: ①

② 右介质

$$\vec{E} = \frac{U}{d} = 5 \times 10^3 V/m$$

$$\vec{D} = \epsilon_{r2} \epsilon_0 \vec{E} = 2.66 \times 10^{-7} V/m$$

$$\vec{D} = \epsilon_{r1} \epsilon_0 \vec{E} = 1.77 \times 10^{-7} V/m$$

(2) $E = 5 \times 10^3 V/m$

轻松

(3) $\vec{P}_1 = \epsilon_0 \chi_{e1} \vec{E} = \epsilon_0 (\epsilon_{r1} - 1) \vec{E} = 1.33 \times 10^{-7} V/m$

$$\vec{P}_2 = \epsilon_0 \chi_{e2} \vec{E} = \epsilon_0 (\epsilon_{r2} - 1) \vec{E} = 2.21 \times 10^{-7} V/m$$

(4) $\sigma_1 = D_1 = 1.77 \times 10^{-7} V/m$ $\sigma_2 = D_2 = 2.66 \times 10^{-7} V/m$

$$\therefore C = \frac{Q}{U} = \frac{\sigma_1 + \sigma_2}{U} S = 2.95 \times 10^{-11} F$$